BRAC UNIVERSITY

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CSE 251

Electronic Devices and Circuits

Lecture 3

Course instructor:

Mir Hamidul Hussain

Lecturer, Department of Computer Science and Engineering,
School of Data and Sciences, BRAC University

Email: ext.hamidul.hussain@bracu.ac.bd

Outline

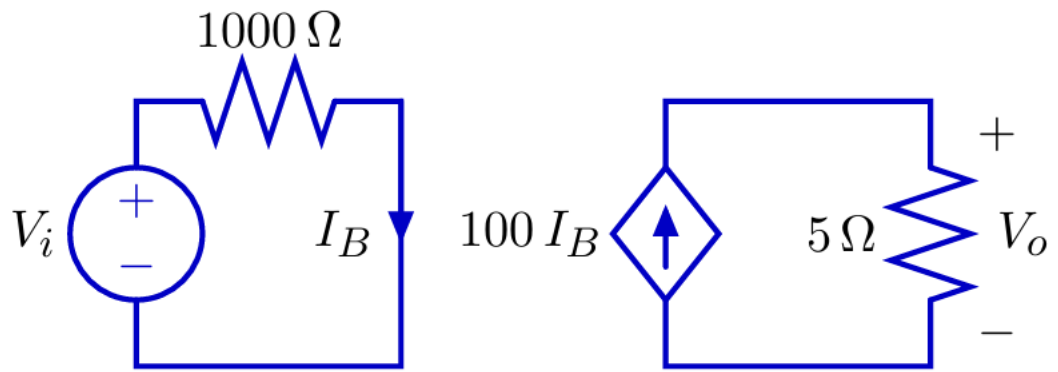
- **Operational Amplifier: Introduction**
  - Dependent Sources
  - Op-Amp: Circuit Symbols and terminal
  - ~~Op-Amp: VTC (Voltage Transfer Characteristics)~~
    - Linear Amplification
    - (Positive and Negative) Saturation
  - Op-Amp: Examples
  - Op-Amp: Physical Entity

Dependent Sources

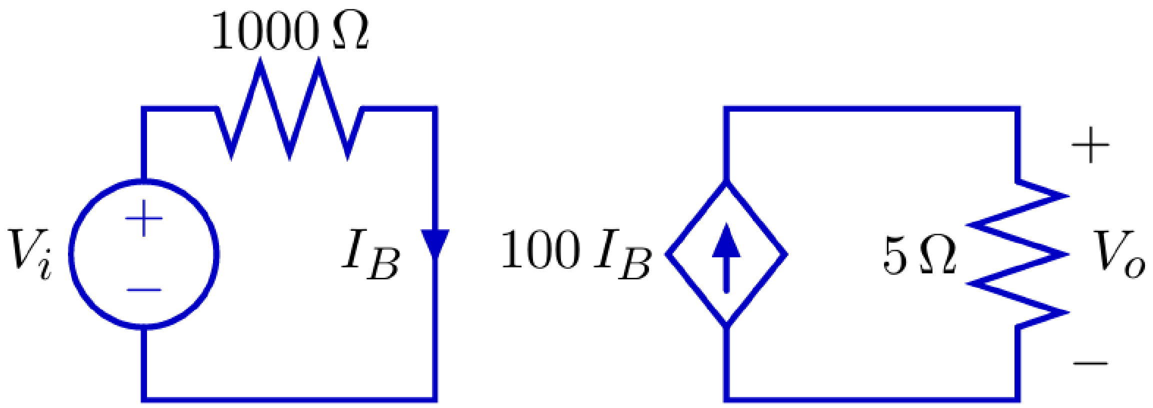
To analyze op-amps, we must understand **dependent source.**

A dependent source generates a voltage or current whose value depends on another voltage or current.

Example: current-controlled current source



Dependent Sources



$$I_B = \frac{V_i}{1000\,\Omega}$$

$$V_o = 100\,I_B \times 5\,\Omega$$

$$= 100\,\frac{V_i}{1000\,\Omega} \times 5\,\Omega = \boxed{\frac{1}{2}V_i}$$

Voltage Gain: $\frac{V_o}{V_i} = \frac{1}{2}$

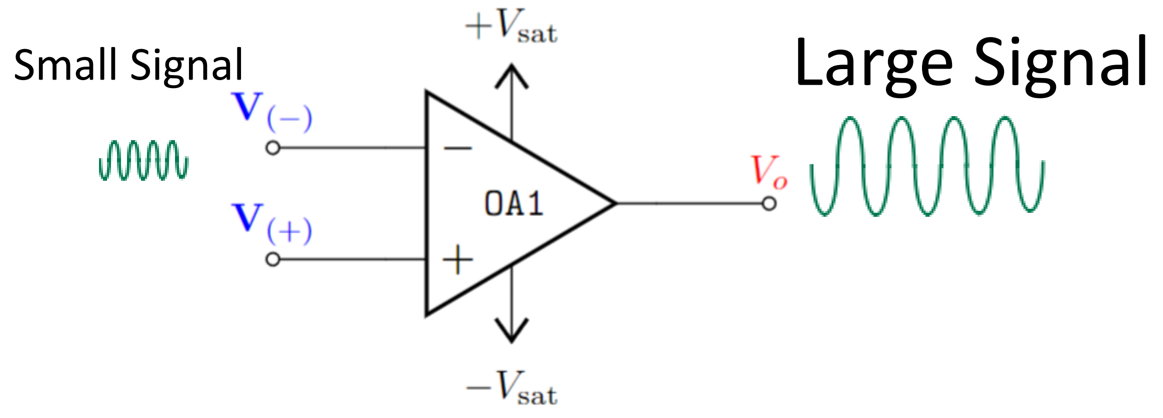
Operational Amplifier: Introduction

- **Operational:**

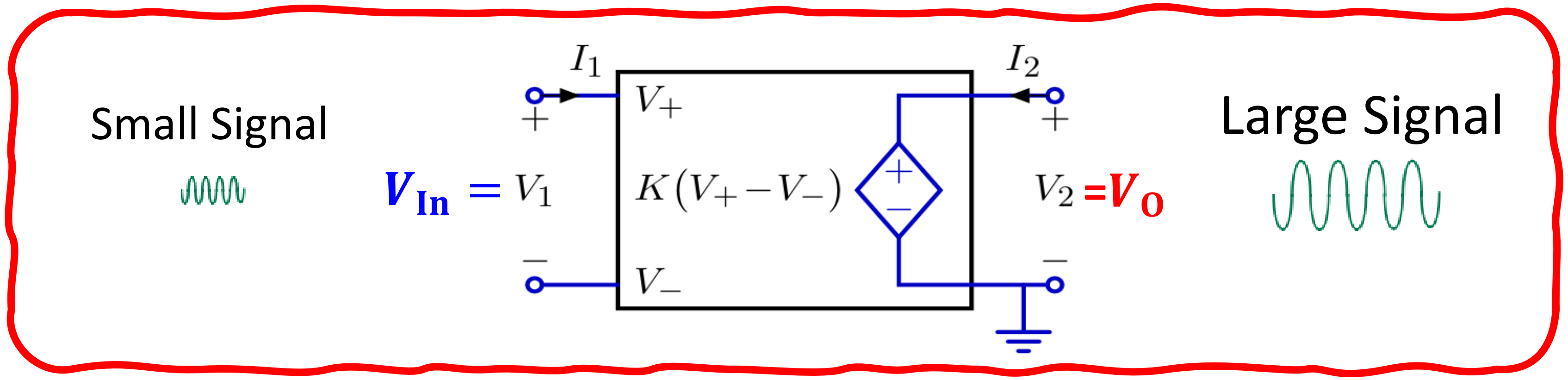
  Mathematical Operations

- **Amplifier:**

  Amplifies input signal/voltage.

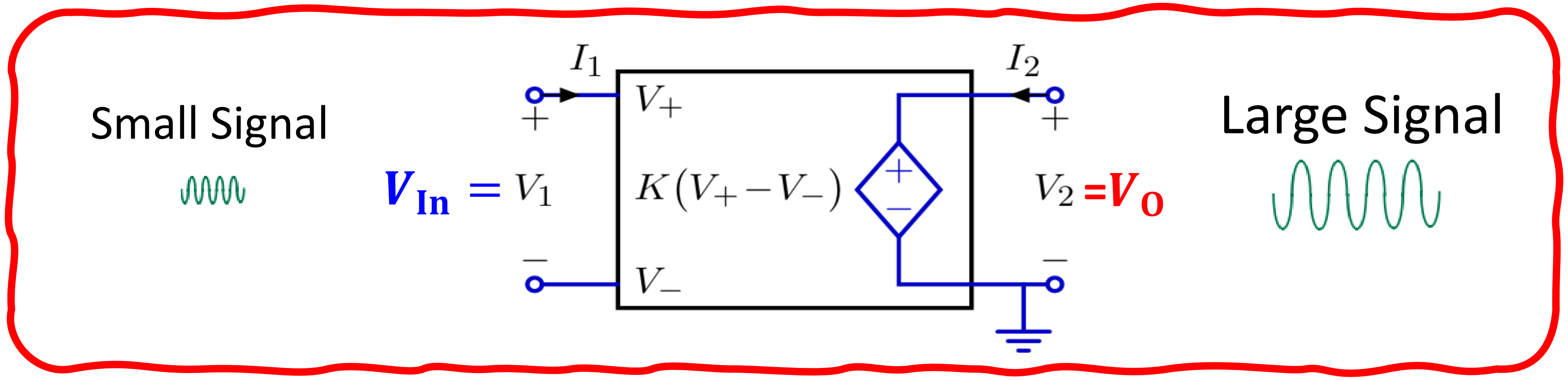


An **op-amp (operational amplifier)** can be modelled by a **voltage- controlled voltage source.**



Operational Amplifier: Introduction

An **op-amp (operational amplifier)** can be represented by a **voltage- controlled voltage source.**

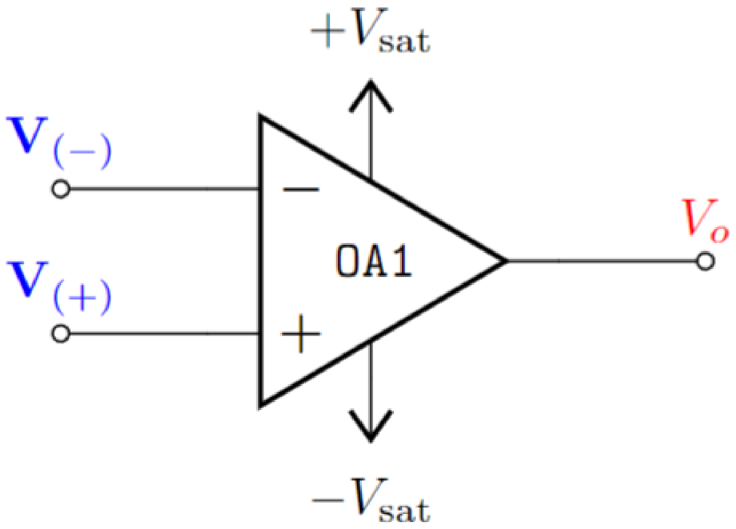


$I_1 = 0$ and $V_O = KV_{In}$ where K is large (typically $K > 10^5$).

Op-Amp: Circuit Symbols and terminal

Positive Saturation
/ Supply Voltage

Inverting
terminal voltage



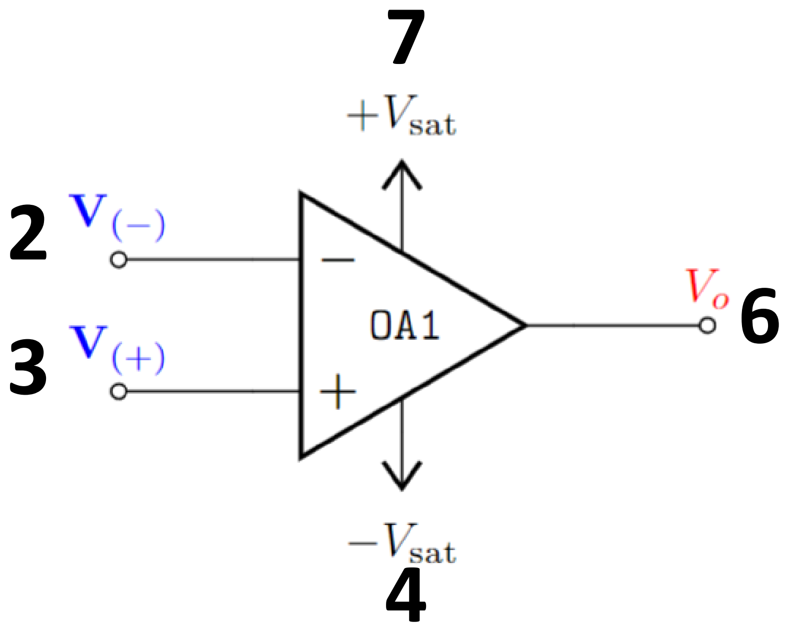
Output Voltage

Non-inverting
terminal voltage

Negative Saturation
/ Supply Voltage

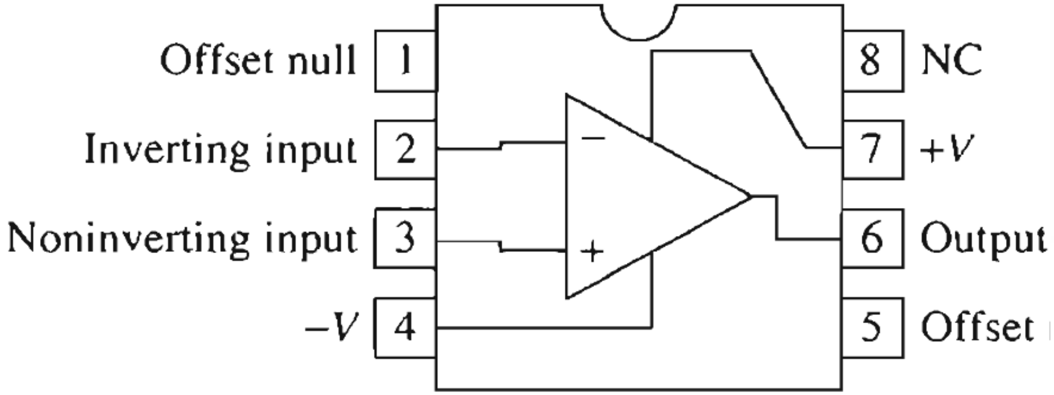
Op-Amp: Physical Entity

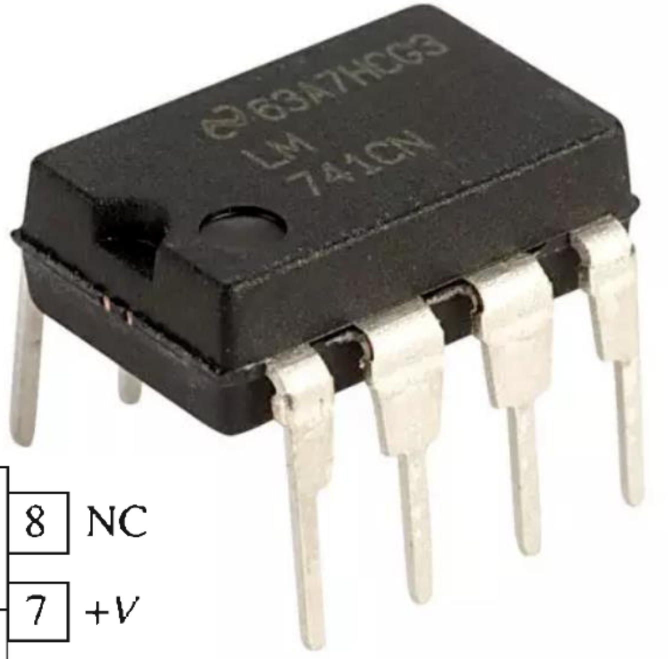
Difference Amplifier – Amplifies the voltage difference between two terminals.



Circuit symbol for the general-purpose op amp. Pin numbering is that for an **8-pin mini-DIP package**

NC – No Connection





μA-741C

C: Commercial ($0° - 70°C$)

I: Industrial ($-25° - 85°C$)

M: Military ($-55° - 125°C$)

Manufacturer **ID**

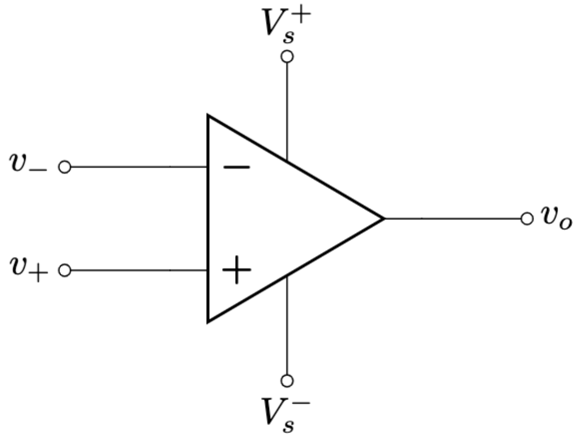
Part Identification Number (PIN)

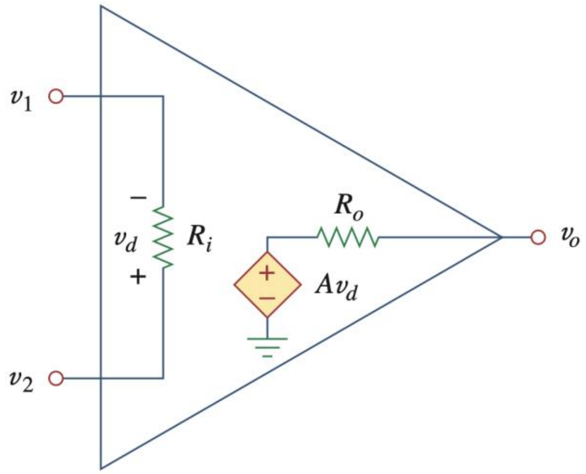
Operational Amplifiers

- An operational amplifier, or **op-amp** for short, is a versatile and powerful integrated circuit that is widely used in a variety of electronic applications.
- An Op-Amp is designed so that it performs some mathematical operations when external components, such as resistors and capacitors, are connected to its terminals.
- The op amp is an electronic device consisting of a complex arrangement of resistors, transistors, capacitors, and diodes. A full discussion of what is inside the op amp is beyond the scope of this course. For now, it will suffice to treat the op amp as a circuit building block and simply study what takes place at its terminals.

Equivalent Circuit

Since op amp is an amplifier, the internal circuit can be modeled using a **voltage controlled voltage source** (VCVS)! (actual circuit is complicated)





$v_1 = v_- =$ voltage of inverting terminal

$v_2 = v_+ =$ voltage of noninverting terminal

$v_d = v_+ - v_- = v_2 - v_1$
$\quad = $ differential input voltage for VCVS

$A =$ Open loop gain

$R_i =$ Input resistance

$R_o =$ Output resistance

The op amp senses the difference between the two inputs, multiplies it by the gain A, and causes the resulting voltage to appear at the output. Thus, the output v_o is given by

$$v_o = Av_d = A(v_2 - v_1) = A(v_+ - v_-)$$

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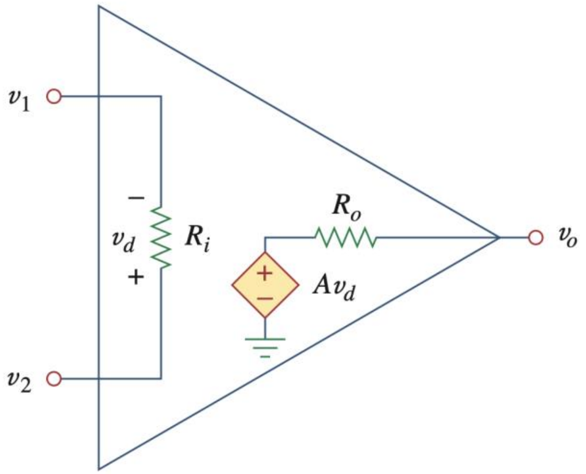

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Parameter	Typical Range

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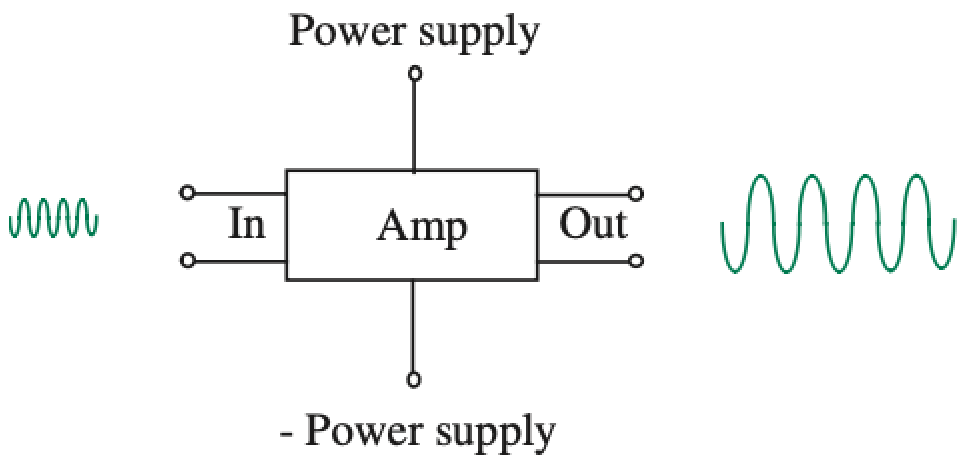
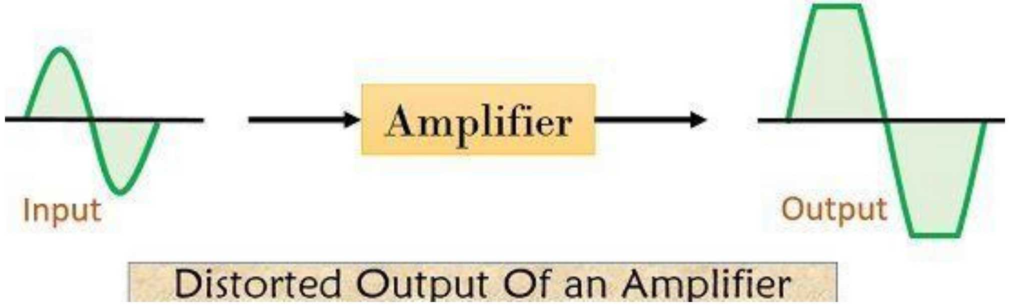
Op-Amp: VTC – Linear Amplification

- **Linear Amplification** only takes place within a **valid input range.**
- Otherwise output will be distorted - - Saturation

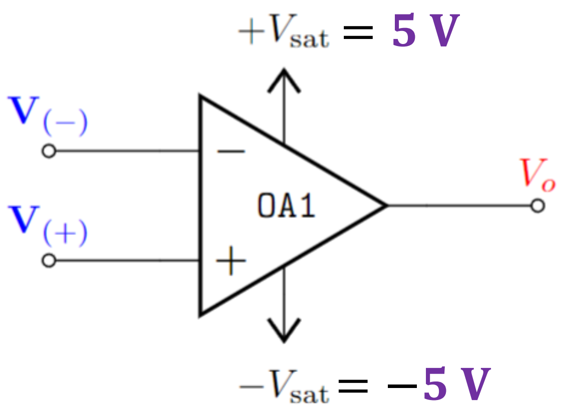




The limiting factor of **linear amplification** is determined by the **power supply** to the amplifier

Op-Amp: VTC - Saturation

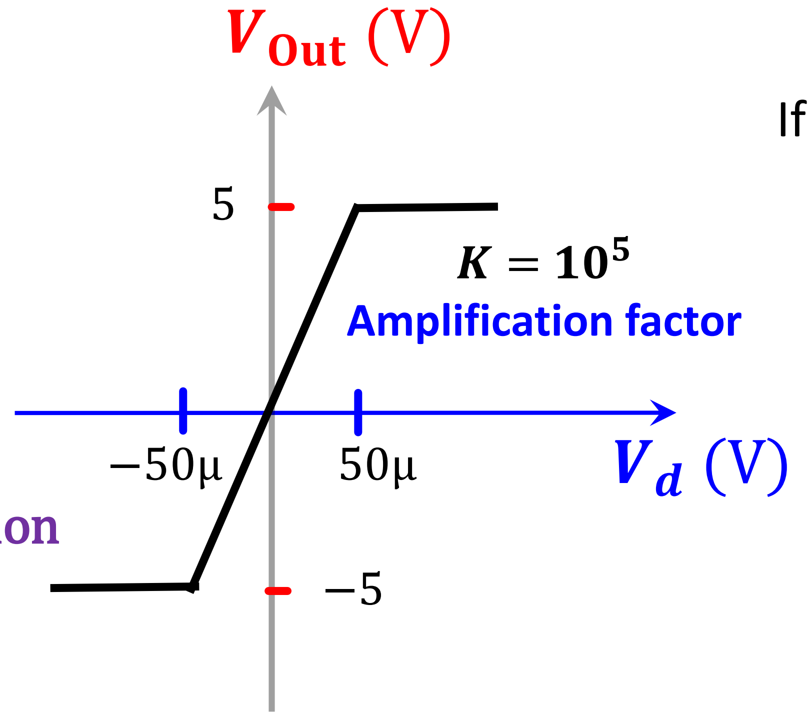
VTC -> Voltage Transfer Characteristics



Non-Linear characteristics

$V_O = K \cdot V_{\text{In}}$: When $-5\ \text{V} < V_O < 5\ \text{V}$

$K \to 10^5$: Gain / Amplification



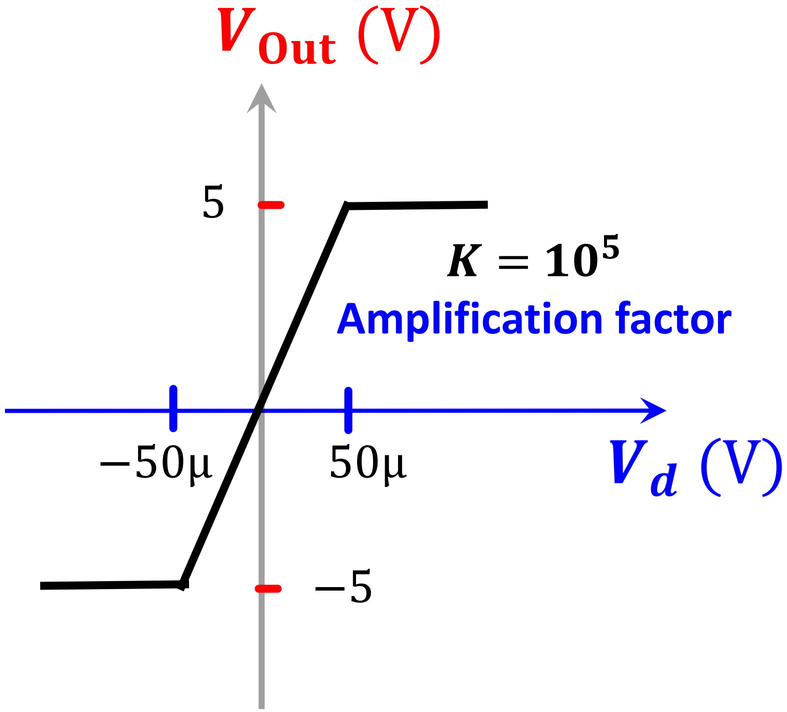
If $V_d > 50\mu$: Positive Saturation

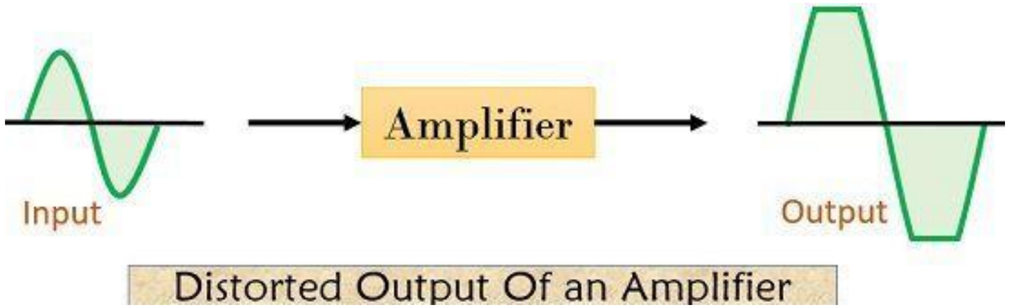
$\Rightarrow V_O = 5$

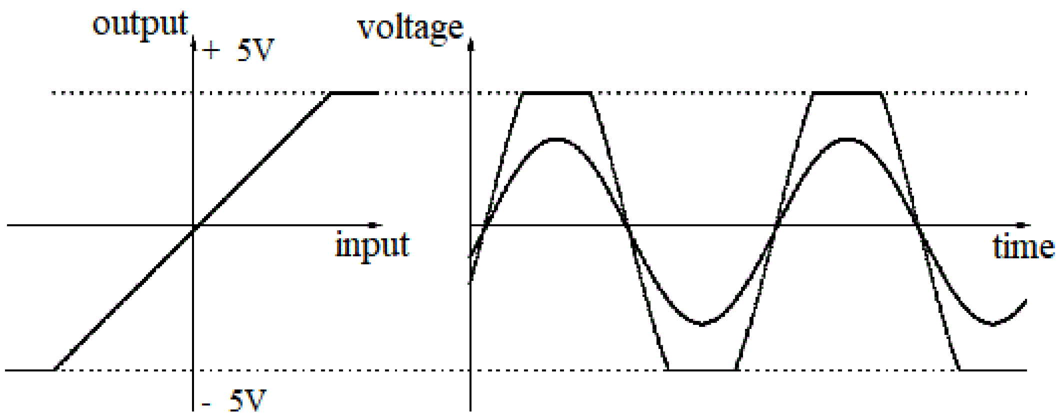
If $V_d < -50\mu$: Negative Saturation

$\Rightarrow V_O = -5$

Op-Amp: VTC - Saturation





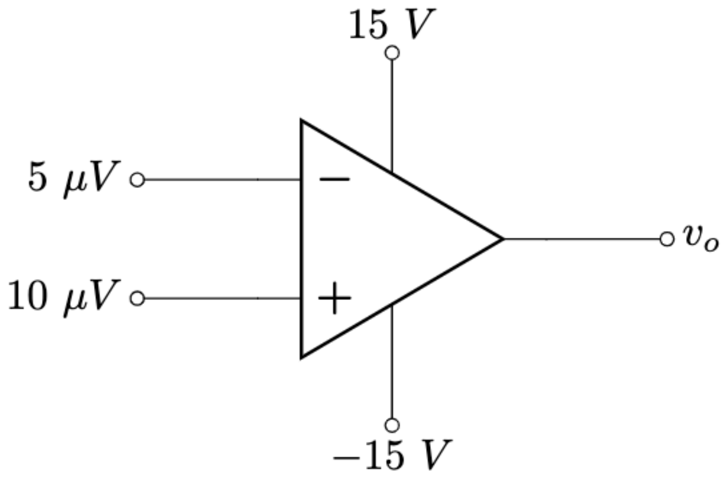


saturation characteristic

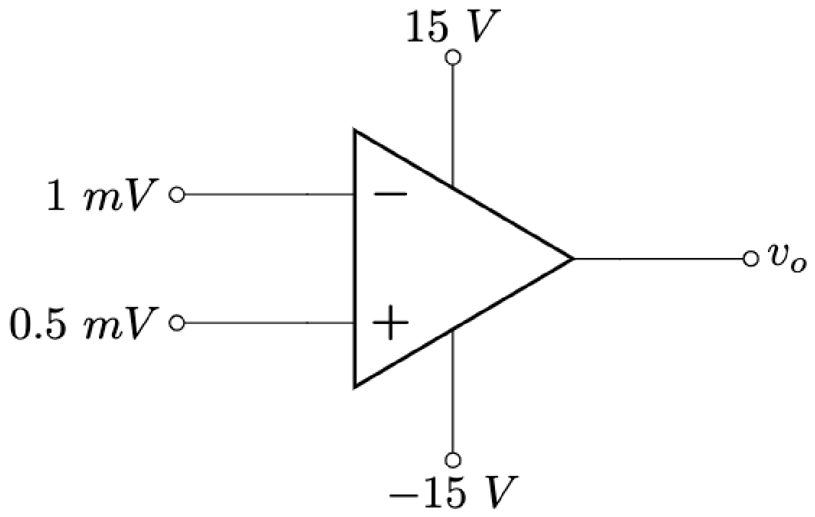
waveform clipping

Op-Amp: Examples

Find v_0



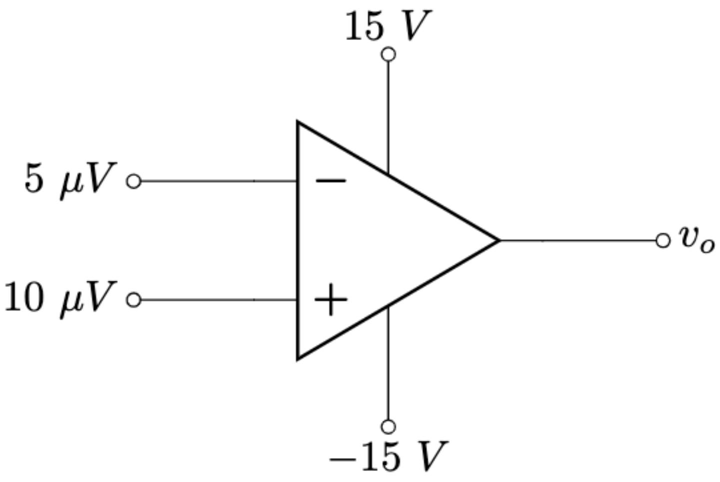
$A = 2 \times 10^5$



$A = 2 \times 10^5$

Example 3

- Find v_o



$A = 2 \times 10^5$

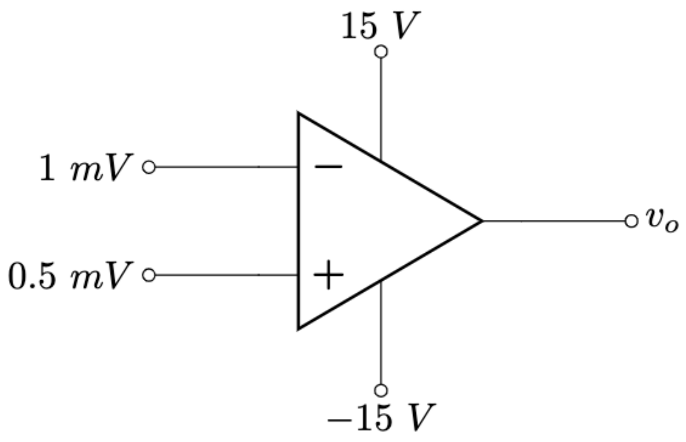
Solution:

$$v_d = v_+ - v_- = 10\ \mu V\ - 5\ \mu V = 5\ \mu V$$
$$\Rightarrow v_o = Av_d = (2\times10^5)\times(5\times10^{-6}) = 1V$$

Since this is in between $-V_{CC}$ and V_{CC}, output valid

Example 4

- Find v_o



$A = 2 \times 10^5$

<u>Solution</u>:

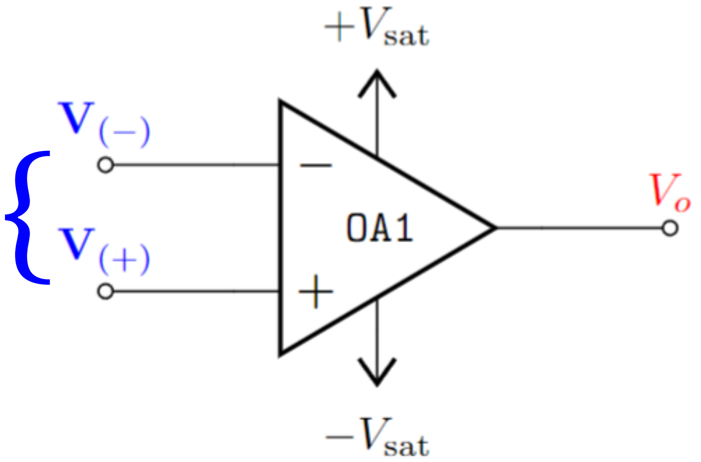
$v_d = v^+ - v^- = 0.5\text{ mV} - 1\text{ mV} = -0.5\text{ mV}$
$\rightarrow v_o = Av_d = (2\text{x}10^5)\text{x}(-0.5\text{x}10^{-3}) = -100\text{V}$

However, the output must be limited within the range of $-V_{CC}$ to $+V_{CC}$. Therefore, the highest output voltage can be -15V.

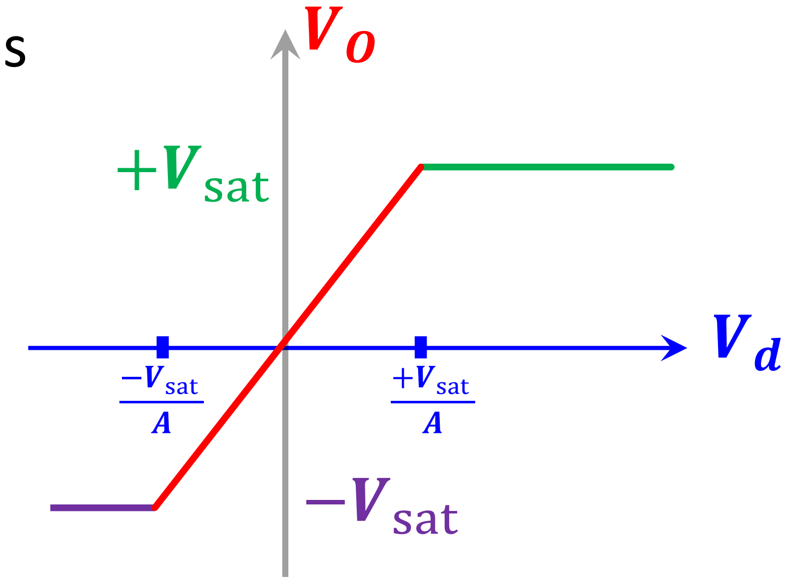
$\therefore v_o = -15\text{V}$

Op-Amp: Summary

Op-Amp **Amplifies** the difference between the voltages at it's two input terminals - $\boldsymbol{V_d}$



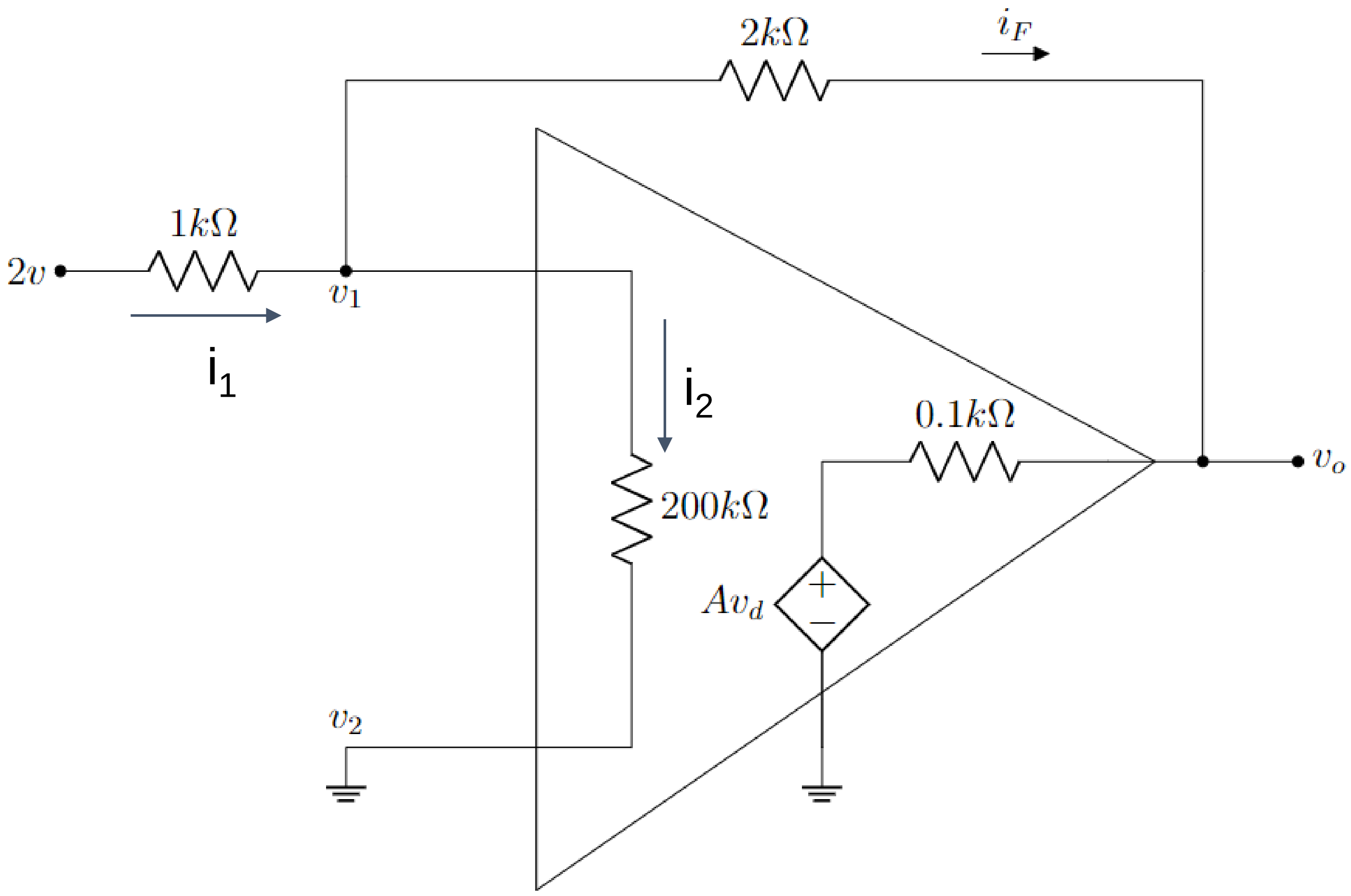
However, the **Amplification** is limited within voltage levels defined by the positive and negative saturation voltages $[-V_{sat}, +V_{sat}]$.



The "ideal" op-amp behaves like a **voltage dependent voltage source** within the linear region.

Example 5

Step 2: Solve using KCL & KVL or nodal



Applying KCL we get,

$$i_1 = i_2 + i_F$$

$\rightarrow$ $(2-v_1)/(1k) = (v_1-v_2)/(200k)+(v_1-v_o)/(2k)$

$\rightarrow$ $(2-v_1)/(1k) = (v_1-0)/(200k)+(v_1-v_o)/(2k)$

$\rightarrow$ $(2-v_1)/(1k) = v_1/(200k)+(v_1-v_o)/(2k)$........(i)

Again, from the figure,

$$i_{0.1k} = i_F$$

$\rightarrow$ $(v_o- Av_d)/(0.1k) = (v_1-v_o)/(2k)$

$\rightarrow$ $(v_o- (2x10^5)x(v_2-v_1))/(0.1k) = (v_1-v_o)/(2k)$

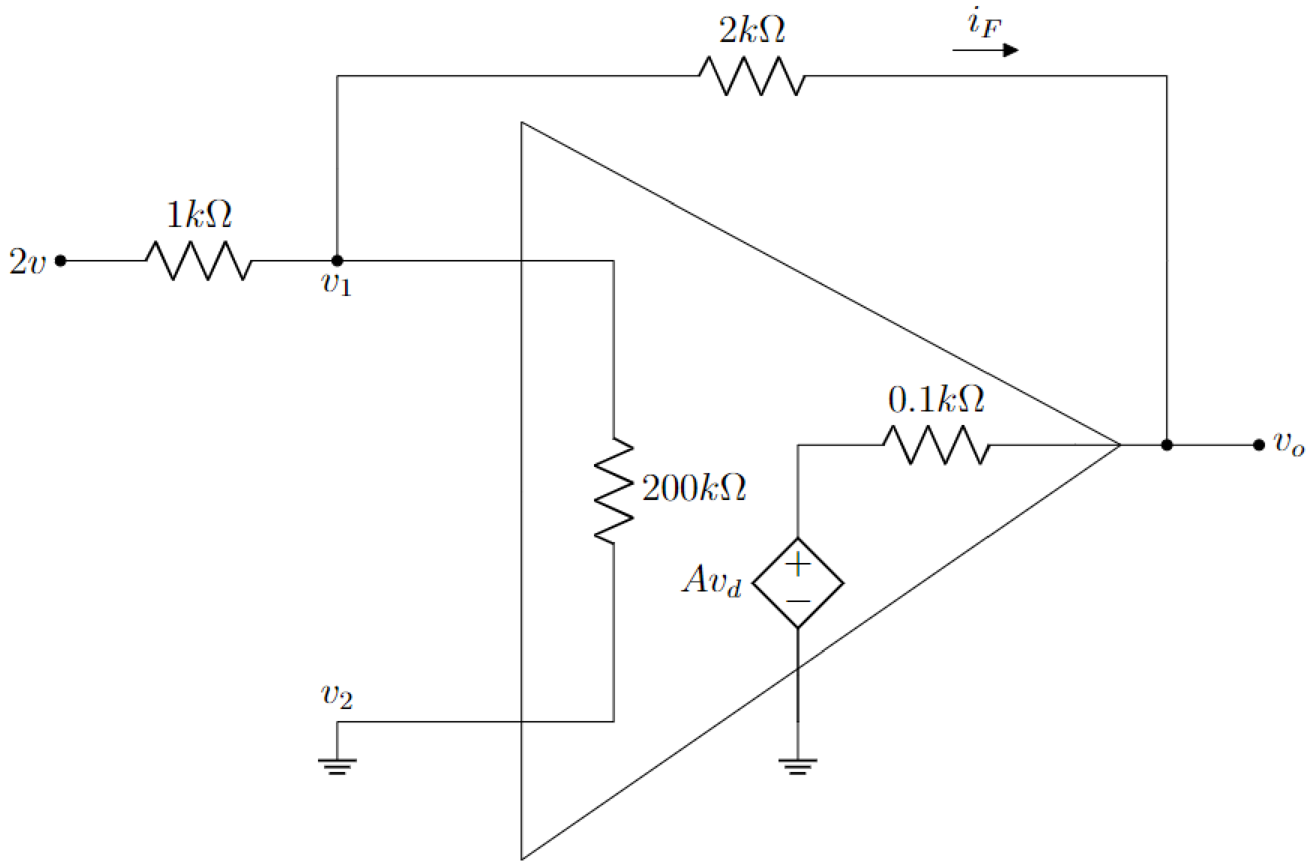
$\rightarrow$ $(v_o- (2x10^5)x(0-v_1))/(0.1k) = (v_1-v_o)/(2k)$

$\rightarrow$ $(v_o- (2x10^5)x(-v_1))/(0.1k) = (v_1-v_o)/(2k)$...(ii)

We can get v_1 and v_o by solving equation (i) and (ii)

Example 5

Step 2: Solve using KCL & KVL or nodal

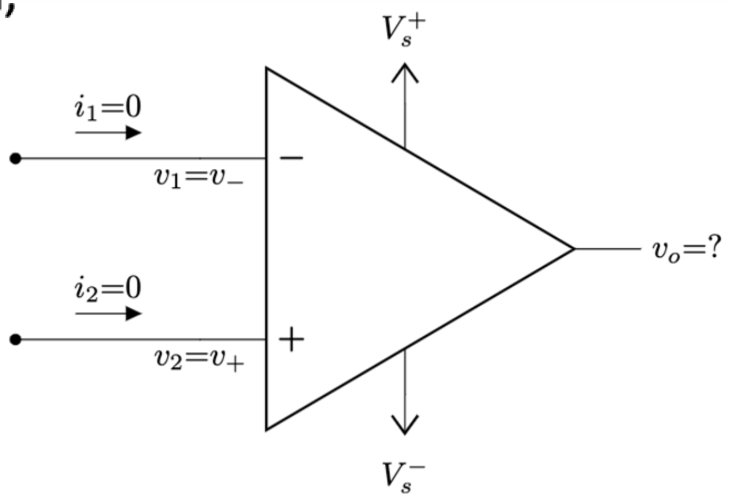


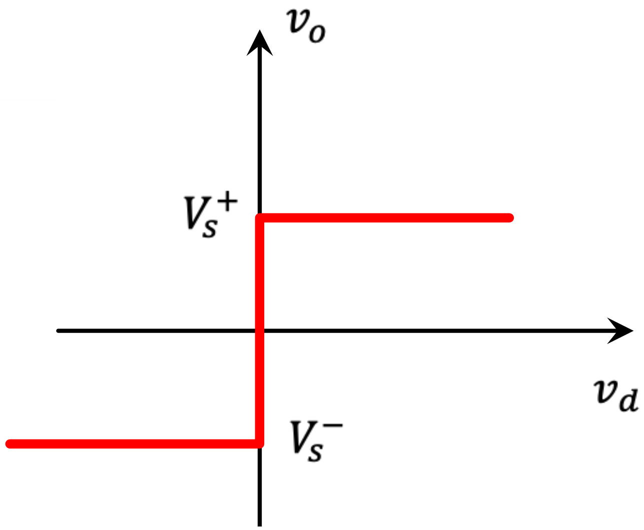
The Ideal Op-Amp

- To facilitate the understanding of op amp circuits, we will assume **ideal op amps**
  - Infinite open-loop gain, $A = \infty$
  - Infinite input resistance, $R_i = \infty = \text{open circuit}$
  - Zero output resistance, $R_o = 0 = \text{short circuit}$
- Although an ideal op amp provides only an approximate analysis, most modern amplifiers have such large gains and input impedances that the approximate analysis is a good one.
- **Circuit solving become much simpler**. As $R_i = \infty$, $i_1 = i_2 = 0$
- Since $A = \infty$, in open-loop configuration, v_o will either be positive saturated or negative saturated (why?)

$$v_o = \begin{cases} V_s^+ \; if \; v_d > 0 \Rightarrow v_2 > v_1 \\ V_s^- \; if \; v_d < 0 \Rightarrow v_2 < v_1 \end{cases}$$

Reminder: $v_d = v_2 - v_1 = v_+ - v_-$

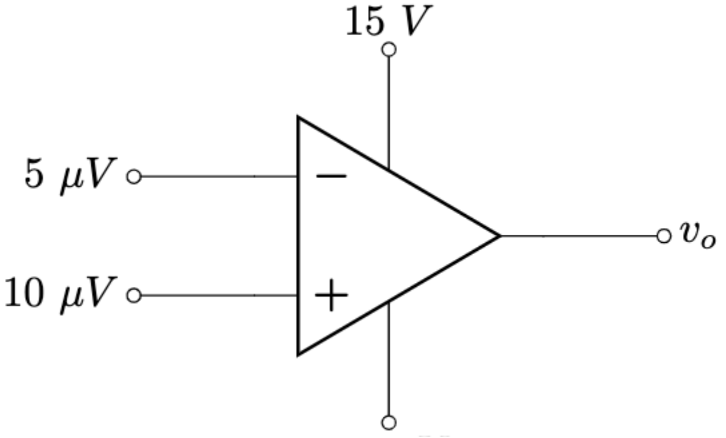




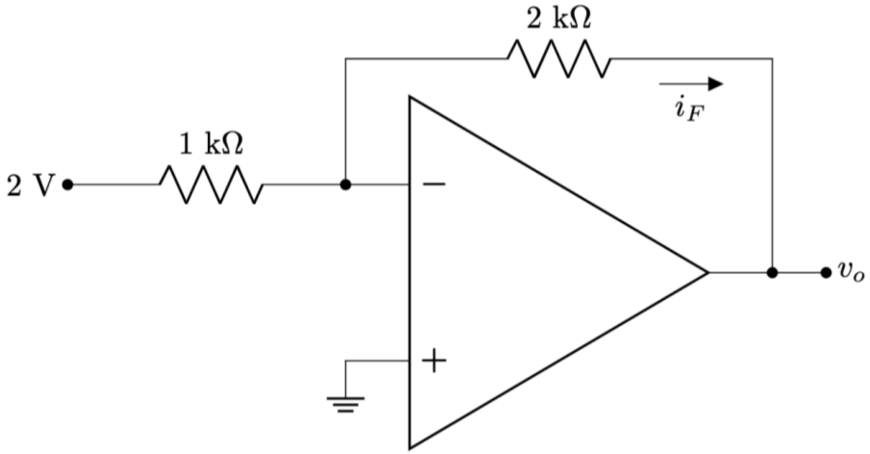
VTC of ideal op amp

Types of Op-Amp configuration

1. Open loop configuration:
   No physical connection between input and output



2. Closed loop configuration:
   Feedback from output terminal



Application - Comparator

- A comparator compares two voltages to determine which is larger.
- The comparator is essentially an op-amp operated in an open-loop configuration
- Two types –
  - (1) **Non-inverting**: outputs a positive voltage ($V_H = V_S^+$) when input is greater than reference
  - (2) **Inverting**: outputs a negative voltage ($V_L = V_S^-$) when input is greater than reference
- Application – smoke detector, turning AC on/off automatically, etc (next lecture)

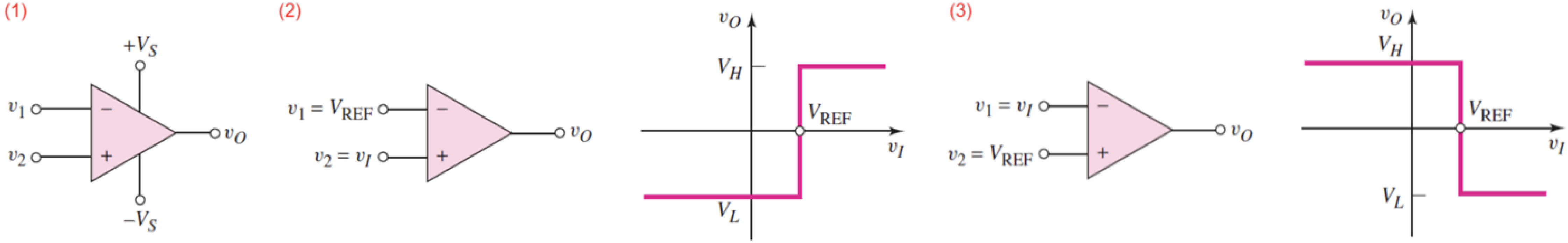


Figure 2: (1) Op-Amp Comparator (2) Noninverting Circuit (3) Inverting Circuit